$$\int \frac{dx}{\sqrt{x^2-3x+2}} = \int \frac{dx}{\sqrt{\left(x-\frac{3}{2}\right)^2-\left(\frac{1}{2}\right)^2}}$$

$$= \log\left|\left(x-\frac{3}{2}\right)+\sqrt{\left(x-\frac{3}{2}\right)^2-\left(\frac{1}{2}\right)^2}\right|+c$$

$$= \log\left|\left(x-\frac{3}{2}\right)+\sqrt{x^2-3x+2}\right|+c$$

Example 2.48

Evaluate $\int \frac{dx}{\sqrt{x^2+25}}$

Solution:

$$\int \frac{dx}{\sqrt{x^2+25}} = \int \frac{dx}{\sqrt{x^2+5^2}}$$

$$= \log\left|x+\sqrt{x^2+5^2}\right|+c \quad = \log\left|x+\sqrt{x^2+25}\right|+c$$

Example 2.49

Evaluate $\int \frac{dx}{\sqrt{x^2+4x+8}}$

Solution:

By completing the squares

$$x^2+4x+8 = (x+2)^2-4+8$$

$$= (x+2)^2+2^2$$

$$\int \frac{dx}{\sqrt{x^2+4x+8}} = \int \frac{dx}{\sqrt{(x+2)^2+2^2}}$$

$$= \log\left|(x+2)+\sqrt{(x+2)^2+2^2}\right|+c$$

$$= \log\left|(x+2)+\sqrt{x^2+4x+8}\right|+c$$

DO YOU KNOW?

$$\int \frac{m}{\sqrt{(mx)^2+a^2}}dx = \int \frac{d(mx)}{\sqrt{(mx)^2+a^2}}$$

$$= \log\left|mx+\sqrt{(mx)^2+a^2}\right|+c$$

Example 2.50

Evaluate $\int \frac{x^3dx}{\sqrt{x^8+1}}$

Solution:

$$\int \frac{x^3dx}{\sqrt{x^8+1}} = \frac{1}{4}\int \frac{4x^3dx}{\sqrt{\left(x^4\right)^2+1^2}}$$

$$= \frac{1}{4}\log\left|x^4+\sqrt{\left(x^4\right)^2+1^2}\right|+c$$

$$= \frac{1}{4}\log\left|x^4+\sqrt{x^8+1}\right|+c$$

Example 2.51

Evaluate $\int \sqrt{x^2-16}\,dx$

Solution:

$$\int \sqrt{x^2-16}\,dx = \int \sqrt{x^2-4^2}\,dx$$

$$= \frac{x}{2}\sqrt{x^2-4^2}-\frac{4^2}{2}\log\left|x+\sqrt{x^2-4^2}\right|+c$$

$$= \frac{x}{2}\sqrt{x^2-16}-8\log\left|x+\sqrt{x^2-16}\right|+c$$

Example 2.52

Evaluate $\int \sqrt{x^2+5}\,dx$

Solution:

$$\int \sqrt{x^2+5}\,dx = \int \sqrt{x^2+\left(\sqrt{5}\right)^2}\,dx$$

$$= \frac{x}{2}\sqrt{x^2+\left(\sqrt{5}\right)^2}+\frac{\left(\sqrt{5}\right)^2}{2}\log\left|x+\sqrt{x^2+\left(\sqrt{5}\right)^2}\right|+c$$

$$= \frac{x}{2}\sqrt{x^2+5}+\frac{5}{2}\log\left|x+\sqrt{x^2+5}\right|+c$$

DO YOU KNOW?

$$\int m\sqrt{(mx)^2+a^2}\,dx = \int \sqrt{(mx)^2+a^2}\,d(mx)$$

$$= \frac{mx}{2}\sqrt{(mx)^2+a^2}+\frac{a^2}{2}\log\left|mx+\sqrt{(mx)^2+a^2}\right|+c$$

Example 2.53

Evaluate $\int \sqrt{4x^2+9}\,dx$

Solution:

$$\int \sqrt{4x^2+9}\,dx$$

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